Exact Rooted Certificates and Sensitivity for Approximation Paving over Arbitrary MASAs

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Abstract

We examine the pinned manuscript on approximation paving over arbitrary maximal abelian subalgebras, including its finite mixed-polynomial argument, buffered pinning, Maharam-slab construction, nontracial expected case, and expectation-free assembly. The source claims self-adjoint strong approximants of norm at most three times the original norm and a universal sixth-power paving count, with error relative to the approximant's own norm. We do not independently establish that global theorem. Our proved contribution is a finite certification layer: cancellation-aware atomic and Hankel realizations preserving invisible characteristic factors; exact zero-insertion-cost and endpoint Schur tests; and dimension-independent, root-gap-free perturbation enclosures for insertion costs at a fixed deleted polynomial. For deterministic descent we reduce the required radius for degree-k diagonal moments to $\lfloor k/2 \rfloor$, sharpen the slab margin and transport coefficient accordingly, and retain a target-weighted transport alternative. A noise-aware polynomial enclosure controls rooted positive mass without inferring an atomic model from approximate rank. We give physical-metric and inexact Schur accounts for Ram coordinates and distinguish these classical constructions from historical novelty. Contribution assessment: C3/5, exact local pinning certificates and a sharper deterministic descent estimate give realized finite gains but do not establish global measurable coloring or improve the paving exponent.

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1 Pinned Problem and Exact Scope

The object of study is OpenAI's *Approximation Paving over Arbitrary Maximal Abelian Subalgebras*, dated September 25, 2026, catalogue item M594, family 300. The catalogue and source pin is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The local source directory and complete read scope appear in Appendix A. The report date is October 7 in the specified local time zone; verification timestamps are recorded in UTC. The contribution scale measures results established here, not the catalogue's applicability level L5.

Let $M \subseteq \mathcal{B}(H)$ be a complex von Neumann algebra acting faithfully, and let $A \subseteq M$ be a unital MASA. Neither separability of H or of the predual nor a normal conditional expectation onto A is assumed. All operators in the main statement are bounded and everywhere defined on H; there is no differential boundary-value problem or implicit boundary condition. For a finite indexed partition $\mathcal{P} = (p_i)_{i=1}^{R}$ by pairwise orthogonal projections in A, with $\sum_i p_i = 1$, define

$$C_{\mathcal{P}}(z) = \sum_{i=1}^{R} p_i z p_i.$$

Zero projections are permitted. The norm is the operator norm in the given representation. The exact quantifier order of the source's Theorem `thm:main` is

$$\exists C_{\mathrm{ap}} > 0 \ \forall \varepsilon \in (0,1) \ \exists R \in \mathbb{N},\ R \le C_{\mathrm{ap}} \varepsilon^{-6} \ \forall (M, A, x, F, \delta), \tag{1.1}$$

where $x = x^* \in M$, $F \subset H$ is finite, and $\delta > 0$, followed by the existence of $y = y^* \in M$, $d = d^* \in A$, and an R-partition such that

$$\|y\| \le 3\,\|x\|\,, \qquad \|(y-x)\xi\| < \delta \quad (\xi \in F), \tag{1.2}$$

$$\|d\| \le \|y\|\,, \qquad \|C_{\mathcal{P}}(y) - d\| \le \varepsilon\,\|y\|\,. \tag{1.3}$$

In particular, R precedes the strong neighborhood. A net indexed by finite vector tests and positive tolerances, not a sequence in general, expresses this conclusion as a strong limit. For $x = 0$ the zero approximant suffices. For $x \neq 0$ rescaling must also rescale δ.

Three non-equivalent assertions. Norm paving of x itself, norm paving of bounded strong approximants y, and a small compression on a projection q with $1 - q$ strongly small are different statements. Popa–Vaes' conversion proposition assumes a normal expectation [5]. The source's separate quadratic corollary invokes a companion result as well as that conversion. The companion manuscript has not been audited here; its quadratic conclusion remains conditional on that separate input. The sixth-power proof does not use it. Neither conversion may silently be applied to an arbitrary expectation-free MASA. We make no claim to a quadratic count for (1.1).

Status of the mathematics. Sections 3–9 contain complete finite proofs. Section 2 follows all substantive source sections and identifies their dependencies; reading and checking a dependency chain is not independent verification of all its global assertions. Our exact finite results do not assume the source's measured-paving theorem or its expectation-free conclusion. Public literature was checked by mathematical title and topic, without sending private corpus material to public services.

2 Stepwise Source Dependency Audit

The source is more than a characteristic-polynomial argument. Its chain has the following distinct stages. Source labels in typewriter font are literal labels in the pinned files, not labels of the present report.

2.1 State topology and the correct quotient

In `sections/preliminaries.tex`, a faithful normal state φ and $A \subseteq M_\varphi$ give a state-preserving normal expectation by Takesaki's theorem. The seminorms are $\|z\|_\varphi^2 = \varphi(z^*z)$ and $\|z\|_\varphi^\# = \|z\|_\varphi + \|z^*\|_\varphi$. Boundedness is essential when these detect strong and strong-star convergence. Decreasing Hilbert-space partition projections yield $C_\mathcal{Q}(z) \to E_A z$ as partitions refine.

The source does not quotient all bounded sequences by state-null sequences. It first defines

$$\mathcal{I}_\varphi = \{(z_j) : \sup_j \|z_j\| < \infty,\ \|z_j\|_\varphi^\# \to 0\},$$

then its two-sided multiplier algebra $\mathcal{C}_\varphi$. Constant sequences and bounded A-sequences are multipliers. Right multiplication by a varying arbitrary bounded element would not satisfy the same state estimate; the centralizer condition is the reason varying partition projections are allowed. Functional calculus in $\mathcal{C}_\varphi/\mathcal{I}_\varphi$ turns a quotient bound into a bounded, self-adjoint, state-null clipping correction. This is not operator-norm convergence of the original sequence.

2.2 Finite polynomials and the initial root bound

The formulas `poly:differential-formula`, `poly:arrow-quotient`, and `poly:sum-local-measures` in `sections/polynomials.tex` are established in Section 3 below with the necessary hypotheses. The source obtains its starting estimate

$$\lambda_{\max}(f_Q) \le 2\sqrt{2/r} + 2/r \tag{2.1}$$

for a zero-diagonal Hermitian contraction and uniform independent colors. Its MSS reduction uses $D_0 = I + T$, columns $D_0^{1/2} e_v \otimes e_{\mathrm{color}(v)}$, covariance sum at most $(2/r)I$, and a positive-defect completion split into columns of squared norm at most one. MSS's normalized covariance root bound, with $d = 2/r$, gives $1 + 2\sqrt{d} + d$ before the shift by -1 [2]. Deleting the added Gram columns is justified through stable principal Gram polynomials and compression interlacing, not by exchanging maximum roots with expectation. The positive bound accommodates the zero roots introduced when passing between U^*U and UU^*.

2.3 Buffered pinning is a cost inequality

Write $h(z) = z + a - \sum_j \nu_j/(z - s_j)$, with $a > 0$, and let

$$m = \int x_+ \, d\gamma(x), \qquad \ell = \big|\{x > 0 : h(x) < 0\}\big|.$$

For children with nonnegative residues and average $\nu_j = \sum_i t_i \nu_{ij}$, the source proves

$$\min_{i : t_i > 0} L(Ka, \nu_i) \le \ell + Cm \tag{2.2}$$

with universal K, C, independent of the number of poles. This coefficient one on ℓ matters: the old insertion cost cancels from the change of total positive-root mass. Replacing ℓ by m is not legitimate.

Here are the intermediate estimates, including the delicate zero cases. Scale to $a = 1$ and put $H = 1 + 2m$. The rooted second-moment and Stieltjes arguments yield

$$\sum_{|s_j| \le 60H} \nu_j \le C_0 H^2, \qquad \sum_{s_j \ge 20H} \frac{\nu_j}{s_j} \le C_0 m, \qquad \sum_{s_j \le -20H} \frac{\nu_j}{|s_j|} \le C_0 H,$$

with the source's permissible $C_0 = 6000$. The positive tail uses a spectral-projection/Bessel estimate integrated in the threshold, not just Markov on a moment record. Boole's one-sided identity has normalization $\sum_j w_j/c$, not twice that number; the two-sided set has twice the measure. For $m \ge 1$, selection of two averaged statistics and a transformed rational sum bound the whole child length by $3(D + 2C_2)m$, with $C_2 = 8C_0$, $D \ge \max(40, 4C_2 + 1)$.

For $m < 1$ the proof instead sets

$$P = \sum_{s_j > 0} \frac{\nu_j}{1 + s_j}, \qquad M_-(u) = \sum_{s_j \le 0} \frac{\nu_j}{u + |s_j|},$$

establishes $P \le C_5 \ell$ and $M_-(\ell/2) \le C_6$, and uses a single choice satisfying three Markov constraints. With $C_5 = 1 + 10C_0$ and $C_6 = \max(21C_0, 1, 28C_5 + 36)$, $K \ge \max(1, 8C_6, 976C_5)$ suffices. Half the parent length is reserved; the remainder is at most $\ell/2 + 8C_0 m$. If $\ell = 0$, the right-limit statistic at zero must be finite; a positive residue at zero is excluded and nonnegative averaging excludes it in every supported child. These details prevent an undefined division at zero. No small practical universal constant or computational speedup is inferred from these conservative estimates.

2.4 Maharam slabs and height coherence

The measured input in `sections/slabs.tex` is a *countable nonsingular Borel* equivalence relation on a standard probability space, principal orbit matrices on $\ell^2([v])$, a bounded-degree supporting graph, and a constant $J \ge 1$ with $J^{-1} \le \omega(v, w) \le J$ on edges. The cocycle convention is $d\mu(w) = \omega(v, w) d\mu(v)$. Thus

$$(v, s) \sim (w, s + \log \omega(v, w)), \qquad d\widetilde{\mu} = d\mu \, e^{-s} ds,$$

is invariant. The weighted matrix is $W = d^{1/2} T d^{1/2}$, $d_{(v,s)} = e^s$, with penalties $a_{(v,s)} = d_{(v,s)} b$ or $d_{(v,s)} K b$. Diagonal-unitary gauge covariance does not change the scalar rooted data. Heights are absolute within a lifted orbit, not reset when the root is changed.

Formal cycle expansions prove finite-radius locality of Laurent coefficients. On each *fixed bounded slab*, compact spectral support gives unique local moment limits and spectral-shift limits; the support bound is not uniform across all slabs. The initial cost vanishes by positivity of every principal polynomial at the tuple b, then a positive-coordinate Taylor expansion for the weighted tuple. Finite-dimensional uniform-color roots alone do not say this without the multiaffine argument.

The high/low distinction in `lem:far-deletion` is essential. High deletions have invariant measure at most e^{-R} but can have huge matrix entries. They are handled by a uniform holomorphic logarithmic-derivative estimate, local coefficient mass transport, and analytic uniqueness on an upper-half-plane domain joined to a fixed-slab neighborhood of infinity. Low deletions cut a matrix of norm $O(e^{-R})$. Central second-moment bounds and the Boole bound

$$\int_M^\infty \xi_h(x) \, dx \le \frac{\sum_j \nu_j}{M}$$

allow analytic estimates to recover integral costs. Uniformity in the partial assignment is necessary for iteration. Our fixed-S sensitivity theorem below does not replace this argument: arbitrary far deletions change the entire deleted system.

For `lem:coherent-choice`, translating heights scales the matrix and the costs by the same e^{α}. Far-deletion comparison makes normalized local tuples nearly invariant in an interior reference band. A Fubini-selected slice and least minimizing color give a measurable base choice. Comparing interior unit bands to that reference controls the error by $C_1 e(R) + C_2(R+1)/L$. Minimizing separately at every lifted point would give a height-dependent coloring, not a coloring of the base.

2.5 Sparse updates and descent

In `sections/pinning.tex`, a polynomial P approximates x_+ through

$$P(-B) = 0, \qquad \int_{-B}^{B} |P'(x) - \mathbf{1}_{(0,\infty)}(x)|\, dx \le \tau.$$

This controls the *single-pin response* by τ because the difference of spectral shifts has absolute value at most one. Uniform approximation of P alone would not provide this dimension-free response estimate. A finite Borel coloring of the distance-$2k$ graph, $k = \max(1, \deg P)$, supplies independent Bernoulli update indicators inside each relevant ball. Projection of a lifted orbit to the base is injective. The probability of two candidates being selected in one ball is $O(\alpha^2)$, giving

$$\mathbb{E}[\Phi_P^{\mathrm{new}} \mid \mathrm{pre}] \le (1 + C\alpha)\Phi_P + \alpha\{\tau_0 + (1 + C)\tau + C_P\alpha\}.$$

The order of parameter choices is S, then τ_0, τ, then L, then P, C_P, finally α; this prevents a circular dependence on the slab. Only finitely many histories are involved at each selected parameter set. Batching deletions with polynomial locality charges the deletion error once per removed vertex, not once per batch.

For descent, a bad assigned radius-k ball forces $\sum_{y \in B_k(u)} m_y/d_y > \sigma$. The valid operator inequality is $A_L \le (A_L)_+$ followed by compression and congruence. Positive part is not asserted to commute with either operation. For $L \ge \max(1, 4k \log J)$, mass transport gives

$$\int ((T_{\mathrm{pav}} - (Kb + \sigma)I)_+^2)_{vv}\, d\mu(v) \le 2\beta + \mu(\mathcal{U}) + \frac{2J^k N_k}{\sigma}\Psi.$$

The remaining unassigned base set gets one additional color. Auxiliary graph colors, polynomial degrees and slab length affect the construction, but not the $r + 1$ final colors. There is no runtime bound in this argument.

2.6 Expected MASAs and the nontracial component

In `sections/expected.tex`, initially assume separable predual and $A \subset M_\varphi$. The algebra N generated by groupoid normalizers is modular invariant, so has E_N, and the decomposition is

$$x - E_A x = Y + Z, \qquad Y = E_N x - E_A x, \quad Z = x - E_N x, \quad \|Y\|, \|Z\| \le 2.$$

The Cartan inclusion $A \subset N$ is represented by a nonsingular measured relation, possibly twisted. Bounded finite-graph approximants to Y, with diagonal removed, have norm at most six. Cocycle cutoffs then make the measured hypotheses hold, with graph and cocycle constants allowed to vary with the approximant. Applying the measured result to $Y/6$ and $-Y/6$ and refining gives $(r+1)^2$ cells and quotient norm at most $6(Kb + \sigma)$.

The included file `sections/expected-atomless.tex` is substantive, not optional. It disintegrates the joint left–right measure of $X\varphi^{1/2}$ and proves that $E_N X = 0$ gives diffuse kernels. An atomic graph need not initially be injective or directly define a normalizer. The source uses the polar intertwiner, an expected type I intermediate algebra with center $\theta(A)$, and an abelian-projection partition to split it into actual normalizing pieces. It then reconstructs the positive L^2 vectors from normal densities over A, contradicting orthogonality to $\overline{N\varphi^{1/2}}$. Atomic central summands of A belong to N and carry no Z component. These are structural steps absent from any finite Hankel test.

Random phases over refining diffuse partitions produce averaged reduced free moments. Nontracial right multiplication requires analytic modular approximants and the bound

$$\|ta\|_\varphi \le \left\|\sigma^\varphi_{i/2}(a)\right\| \|t\|_\varphi .$$

The source replaces fixed factors from right to left so that already replaced factors have controlled right actions. Bounded differences give variance bounded by a word constant times $\sum_q \varphi(q)^2 \to 0$, allowing deterministic labels for countably many words. It does not simply invoke trace cyclicity.

The two-by-two symmetry dilation uses a free diffuse second diagonal algebra, because the ordinary second copy of A could leave an atomic cross-kernel even when $z = 0$. With $z = Z/2$ and $s = (1 - z^2)^{1/2}$,

$$U = \begin{pmatrix} z & s \\ s & -z \end{pmatrix}, \qquad D = A \oplus B_0.$$

Free compression bounds its limit by $2/\sqrt{n}$. Moments imply state-null spectral excess, not norm convergence. Passing through the multiplier quotient and the first matrix corner gives $4/\sqrt{n}$ for Z. Refinement with the normalizer partition yields $R_{\mathrm{st}}(\gamma) = (r+1)^2 n$, and clipping produces self-adjoint corrections w_j with $\|w_j\| \le 2$ and $\|w_j\|_\varphi \to 0$.

2.7 Nonseparable and expectation-free assembly

The file `sections/general.tex` first builds a countably generated subalgebra closed under the relevant expectations and including the partition witnesses. This ensures that its intersection with A is still a MASA; taking an arbitrary separable subalgebra would not do so. The bound R_{st} is unchanged.

Next let $e \in A$ be the supremum of supports of A-central normal positive functionals, $h = 1 - e$. The source constructs a faithful normal expectation on eMe by orthogonal A-corners, possibly uncountably many. Neither e nor h is asserted central in M. On hMh there are no nonzero centralizing normal positive functionals. Standard-form partitions make conjugate normal states almost disjoint; pairwise separating projections are converted to sequential positive effects, not incorrectly summed as if orthogonal. Their cyclic average gives a positive contraction s with $p_i s p_i \le p_i/m$ and sxs strongly near x on the prescribed tests and their images.

Only after choosing finite vectors is $e = f + g$ split so that fMf has a faithful centralizing state and g annihilates all test vectors. The three diagonal corners are replaced; every off-diagonal block of x between f, g, h is retained until the final partition. Thus $y - x$ is block diagonal of norm at most two and $\|y\| \le 3$ for $\|x\| = 1$. A unit norming test vector with $\|x\xi_0\| > 3/4$, approximated to accuracy below $1/4$, ensures $\|y\| > 1/2$. Clipping the diagonal witness to $[-\|y\|, \|y\|]$ doubles the absolute residual $\gamma = \varepsilon/4$, which is still at most $\varepsilon\|y\|$. This accounts for the approximant-relative normalization in (1.3).

The source's explicit choices are

$$b = \sigma = \frac{\varepsilon}{96(K+1)}, \quad r = \left\lceil \frac{32}{b^2} \right\rceil, \quad n = \left\lceil \frac{1024}{\varepsilon^2} \right\rceil, \quad m = \left\lceil \frac{4}{\varepsilon} \right\rceil,$$

and $R = (r+1)^2 n + m + 1$. This is the exact origin of the sixth power. A finite improvement of a residue test does not remove either refinement or any structural assumption used in this chain.

3 Finite Polynomial and Spectral Identities

Fix a finite set Q, $|Q| = N$, $W = W^* \in \mathbb{C}^{N\times N}$ with $W_{vv} = 0$, real a_v, and independent colors in $\{1, \dots, r\}$ with probabilities $t_{vi} \geq 0$, $\sum_i t_{vi} = 1$. Set

$$f_Q(Y) = \mathbb{E}\det(\mathrm{diag}(Y_v + a_v) - W_{\mathrm{pav}}), \qquad f_\varnothing = 1. \tag{3.1}$$

All principal restrictions carry the restricted probabilities and penalties. In particular, $\partial_{Y_v} f_Q = f_{Q\setminus v}$. The univariate symbol $f_Q(z)$ means that *all* remaining diagonal variables equal z; its derivative is the sum of the separate deletion polynomials.

Proposition 3.1 (Normalized differential identity)**.** *With* $T_i^{1/2} = \mathrm{diag}(\sqrt{t_{vi}})$,

$$f_Q(Y) = \left(\prod_{v\in Q} \frac{\partial_{Y_v}^{r-1}}{r!}\right) \prod_{i=1}^{r} \det(\mathrm{diag}(Y + a) - rT_i^{1/2} W T_i^{1/2}). \tag{3.2}$$

The polynomial is upper-half-plane stable and $f_Q(z)$ is real-rooted. The algebraic identity remains valid for complex probabilities summing to one, complex penalties and complex zero-diagonal W; the stability conclusion in general does not.

Proof. In a determinant expansion a nontrivial directed permutation cycle C has combined permutation/entry sign minus. In the expectation its factor is

$$-\prod_{(v,w)\in C} W_{vw} \sum_i \prod_{v\in C} t_{vi}.$$

Disjoint cycles use independent vertices. In the determinant product, if a vertex belongs to cycles in k factors, its remaining power of $Y_v + a_v$ is $r - k$. The derivative annihilates it for $k \geq 2$. For $k = 0$ its multiplier is one and leaves $Y_v + a_v$; for $k = 1$ its multiplier is $1/r$, canceling the factor r at that cycle vertex. This proves the identity, including the exact $r!$ and complex extension. The two square roots at a cycle vertex multiply to t_{vi}, independently of branch choices used consistently there.

For real data, each determinant is nonzero when all $\mathrm{Im}\, Y_v > 0$: a null vector contradicts the strictly positive imaginary part of the quadratic form. Products preserve stability. Differentiating one coordinate preserves half-plane zero-freeness by the univariate Gauss–Lucas theorem; it cannot yield the zero polynomial in this process because its highest-coordinate coefficient is the corresponding already-processed principal product, nonzero by induction on vertices. Repeating $r-1$ times at each vertex proves stability. Real coefficients and conjugation exclude nonreal roots of the univariate specialization. For $r = 1$ the differential operator is the identity. Zero probabilities cause no difficulty. Zero diagonal is essential for the displayed cycle calculation; a deterministic nonzero diagonal should first be absorbed into the penalties, not treated as a probabilistically thinned loop. □

Proposition 3.2 (Rooted arrow representation)**.** *Fix v, set $f = f_Q(z)$ and $g = f_{Q\setminus v}(z)$. If the roots of g, including multiplicities, are $s_1, \dots, s_{N-1}$, there are $\nu_j \geq 0$ such that*

$$h(z) = \frac{f(z)}{g(z)} = z + a_v - \sum_j \frac{\nu_j}{z - s_j}, \tag{3.3}$$

$$B = \begin{pmatrix} -a_v & c^* \\ c & S \end{pmatrix}, \quad S = \mathrm{diag}(s_j), \quad c_j = \sqrt{\nu_j}, \quad \det(zI - B) = f(z). \tag{3.4}$$

For $e = (1, 0, \dots, 0)^T$, $G = g/f = e^*(zI - B)^{-1}e$ is the Stieltjes transform of a probability measure γ. Its first moments are

$$\mu_0 = 1, \quad \mu_1 = -a_v, \quad \mu_2 = a_v^2 + \sum_j \nu_j, \qquad \sum_j \nu_j = \sum_{w \neq v} |W_{vw}|^2 \sum_i t_{vi} t_{wi}. \tag{3.5}$$

Proof. Fix the other diagonal variables at an upper-half-plane z. Division by $g(z) \neq 0$ leaves a monic affine polynomial in Y_v with no upper-half-plane zero. Evaluating at $Y_v = z$ yields $\operatorname{Im} h(z) \geq \operatorname{Im} z$. The poles of the rational real function h are real. A pole of order at least two would give both signs of the imaginary part along upper rays near that pole; a simple polar residue must be nonpositive by vertical approach. Its polynomial part is $z + a_v$, from the two leading coefficients of f, g. This gives (3.3). At a repeated pole one may assign the total mass to one copy and zero to the other copies. The Schur complement proves the determinant and resolvent identities, with extension across canceled factors as polynomial or rational identities. The spectral theorem gives the positive measure and its first moments. The z^{-1} term of f/g is minus the sum of the length-two cycle weights through v, proving the last formula. □

Let $N_f(x)$ count roots strictly greater than x, with multiplicities, and define $\xi = N_f - N_g$. Schur-complement inertia shows $\xi(x) = \mathbf{1}_{\{h(x)<0\}}$ off a finite set, hence $0 \leq \xi \leq 1$ almost everywhere. Put

$$\mathcal{E}(f) = \sum_{f(\lambda)=0} \lambda_+, \qquad \ell = \mathcal{E}(f) - \mathcal{E}(g) = \int_0^\infty \xi(x)\, dx, \qquad m = e^* B_+ e = \int x_+ \, d\gamma(x). \tag{3.6}$$

The distinction is strict even for a two-by-two arrow. At all vertices, the deletion chain rule gives $\sum_v G_v = f_Q'/f_Q$, hence

$$\sum_v \gamma_v = \sum_{f_Q(\lambda)=0} \delta_\lambda, \qquad \sum_v m_v = \mathcal{E}(f_Q). \tag{3.7}$$

These are costs of an *expected polynomial*. They are not generally expectations of matrix positive-part traces.

Pinning the color at v leaves g unchanged. Thus $h = \sum_i t_{vi} h_i$ and, at each distinct pole, $\nu_j = \sum_i t_{vi} \nu_{ij}$. Increasing a_v to $K a_v$ adds $(K-1)a_v$ to h_i. In what follows, residues at repeated copies are always placed consistently across all alternatives. This common-pole realization is what makes a local perturbation a change in a single arrow row and column.

4 Multiplicity-Aware Moment Certificates

An arrow matrix realizes the full characteristic polynomial; its rooted measure realizes only its cyclic part. Confusing the two loses multiplicities or uncoupled eigenvalues. The following gives an exact finite certificate, using the Hankel and atomic-support logic of the actual private passages listed in Appendix B.

Theorem 4.1 (Cyclic realization with the canceled factor retained)**.** *For the pair f, g in Proposition 3.2, let $d = \gcd(f, g)$ be monic and write $f = dp$, $g = dq$, $\deg p = k$, $\deg q = k - 1$. Then p has k distinct real roots λ_j and*

$$G(z) = \frac{q(z)}{p(z)} = \sum_{j=1}^{k} \frac{w_j}{z - \lambda_j}, \qquad w_j = \frac{q(\lambda_j)}{p'(\lambda_j)} > 0, \qquad \sum_j w_j = 1. \tag{4.1}$$

With $\mu_l = \sum_j w_j \lambda_j^l$, define

$$H = (\mu_{i+j})_{i,j=0}^{k-1}, \qquad L = (\mu_{i+j+1})_{i,j=0}^{k-1}, \qquad C = H^{-1}L.$$

Then $H > 0$, $HC = C^*H$, and

$$\det(zH - L) = \det(H)p(z), \tag{4.2}$$

$$e_0^* H(zI - C)^{-1} e_0 = G(z), \tag{4.3}$$

$$H_k = (\mu_{i+j})_{i,j=0}^{k} \succeq 0, \quad \operatorname{rank} H_k = k, \quad H_k \operatorname{coeff}(p) = 0. \tag{4.4}$$

Here coefficients are in increasing power order. The data through degree $2k$ uniquely determine γ among positive real measures; they do not determine the canceled polynomial d. The full spectral record must retain $f = dp$ and $g = dq$. Nevertheless,

$$m = \sum_j w_j (\lambda_j)_+, \qquad \ell = \mathcal{E}(p) - \mathcal{E}(q), \tag{4.5}$$

where roots of q are counted with multiplicity.

Proof. The spectral theorem makes G a finite sum of positive atoms over the eigenvalues with nonzero e-projection. In reduced form its poles are simple, exactly the roots of p, and all its residues are positive. Its $1/z$ asymptotic is one, giving their sum. Define $J : \mathbb{C}^k \to \mathbb{C}^N$ by $Jc = \sum_{i=0}^{k-1} c_i B^i e$. Then $J^*J = H$, $J^*BJ = L$. For a polynomial u of degree below k,

$$c^* H c = \sum_j w_j |u(\lambda_j)|^2.$$

This vanishes only if $u = 0$, by the distinctness of k nodes. Thus $H > 0$. The range of J is the k-dimensional cyclic subspace of B at e; it is invariant, since $p(B)e = 0$. Therefore $BJ = JC$, not just $J^*BJ = HC$. Its spectrum consists of the k distinct nodes, proving (4.2); $Je_0 = e$ and $BJ = JC$ prove (4.3). Gram factorization gives (4.4).

For any other positive measure with the stated $2k + 1$ moments, the integral of $|p|^2$ is zero. Positivity forces support in the finite real zero set of p. The first k moment equations uniquely recover its weights by an invertible Vandermonde matrix. Existence is provided by (4.1), so this proves uniqueness. Multiplying both f and g by a monic real-rooted factor does not change G, demonstrating the missing information. Finally the common roots cancel from the difference of positive-root sums, proving (4.5). □

Certificate contract. For proposed rational or algebraic input f, g, check degrees, leading coefficients, the exact gcd, real-root isolation of p, positivity of (4.1), and the identity $q/p = \sum_j w_j/(z - \lambda_j)$ after clearing denominators. The monic-normalization check ensures total mass one. Compute the moments and verify the entire supplied record and (4.4); do not infer exact rank by rounding a small singular value to zero. If only moments are supplied, additionally verify a candidate monic annihilator, its distinct real roots in the declared support, positive recovered weights, and reproduction of *every* supplied moment. This is the complete-check construction in the volume 21 passage, not an assertion that an arbitrary Hankel positive-semidefinite test solves a truncated moment problem.

Example 4.2 (Repeated poles and invisible positive roots)**.** Take $a = 1$, $S = \operatorname{diag}(-2, -2, 4)$ and $c = (\sqrt{6}, 0, 0)^T$. Then

$$h(z) = z + 1 - \frac{6}{z+2}, \quad f(z) = (z+2)(z-4)(z-1)(z+4), \quad g(z) = (z+2)^2(z-4).$$

Thus $d = (z+2)(z-4)$, $p = (z-1)(z+4)$, $q = z+2$, and

$$\gamma = \tfrac{3}{5}\delta_1 + \tfrac{2}{5}\delta_{-4}, \qquad m = \tfrac{3}{5}, \quad \ell = 1, \quad \mathcal{E}(f) = 5, \quad \mathcal{E}(g) = 4.$$

The positive root 4 has zero rooted weight and must not be erased from f. The moment record begins $(1, -1, 7, -25, 103)$ and

$$H = \begin{pmatrix} 1 & -1 \\ -1 & 7 \end{pmatrix}, \quad L = \begin{pmatrix} -1 & 7 \\ 7 & -25 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 4 \\ 1 & -3 \end{pmatrix}, \quad \det H = 6.$$

The non-Hermitian Euclidean appearance of C is harmless because $HC = C^*H$. Removing d changes full root counts but changes neither m nor ℓ.

5 Exact Zero-Cost and Threshold Tests

The source's universal buffer controls arbitrary configurations. A given finite configuration admits a sharper exact decision, including singular endpoints and positive invisible roots.

Theorem 5.1 (Zero insertion cost, including uncoupled poles)**.** *Let $a > 0$, $s_j \in \mathbb{R}$, $\nu_j \geq 0$, and form (3.3)–(3.4) without any paving assumption. Define m, ℓ by (3.6). The following are equivalent:*

(i) $\ell = 0$*;*

(ii) $m = 0$*;*

(iii) every residue at $s_j \geq 0$ is zero and $\sum_{s_j<0} \nu_j/(-s_j) \leq a$.

The condition permits positive or repeated poles with zero residues. For a child obtained by replacing the positive penalty by Ka, its zero-cost threshold, among $K \geq 1$, is exactly

$$K_{\text{zero}} = \max\left(1, \frac{1}{a}\sum_{s_j<0} \frac{\nu_{ij}}{-s_j}\right) \tag{5.1}$$

if every nonnegative-pole residue vanishes; otherwise no finite penalty gives zero insertion cost.

Proof. If a positive pole carries mass, immediately to its right h is negative on a positive-length interval. A mass at zero likewise makes $h(x) \to -\infty$ as $x \downarrow 0$. Both contradict $\ell = 0$. With these masses excluded, h is continuous and strictly increasing on $(0, \infty)$, since

$$h'(x) = 1 + \sum_{s_j<0} \frac{\nu_j}{(x - s_j)^2} \geq 1.$$

Its right limit is $a - \sum_{s_j<0} \nu_j/(-s_j)$. A negative limit produces a positive-length negative interval, whereas a nonnegative limit gives none. This proves (i)⇔(iii), including equality at zero.

For (iii)⇒(ii), discard the uncoupled coordinates and use $S < 0$ on the remaining ones. The negative-semidefinite Schur criterion at zero says that the remaining arrow is nonpositive precisely when the displayed sum is at most a. Its rooted spectral measure then has no positive support. For (ii)⇒(i), reduce f, g as in Theorem 4.1. All roots of p are in $(-\infty, 0]$. The numerator q has only real zeros between consecutive support nodes: on a gap the positive-residue function G strictly decreases from $+\infty$ to $-\infty$, with one simple zero, and it has no zero outside the extreme support nodes. For $k = 1$, $q = 1$. Thus $\mathcal{E}(p) = \mathcal{E}(q) = 0$, so $\ell = 0$. Replacing a by Ka proves (5.1). □

Corollary 5.2 (Zero parent cost needs no buffer at one finite pin)**.** *If the parent has $m = 0$ and $\nu_j = \sum_i t_i \nu_{ij}$ with $t_i \geq 0$, $\sum_i t_i = 1$, then some i with $t_i > 0$ has zero child insertion cost with $K = 1$.*

Proof. Every supported child has zero residues at nonnegative poles by nonnegative averaging. The weighted average of their negative-pole Schur sums is the parent sum, at most a. At least one supported child has sum at most a. Apply Theorem 5.1. □

This is an exact finite one-step strengthening in the zero-cost case, not a measurable coloring theorem. Even iteration on one finite matrix does not make the choices coherent on Maharam fibers.

Proposition 5.3 (Full endpoint threshold certificate)**.** *For any real θ, including a pole of the deleted polynomial,*

$$B \preceq \theta I \quad \Longleftrightarrow \quad D_\theta := \theta I - S \succeq 0, \quad c \in \operatorname{ran} D_\theta, \quad \theta + a - c^* D_\theta^\dagger c \geq 0. \tag{5.2}$$

If $\theta > \max_j s_j$, this is exactly $h(\theta) \geq 0$. If $\theta = s_j$, the range condition requires zero coupling to every such coordinate. The statement concerns the full *matrix; an invisible eigenvalue above θ still violates it.*

Proof. The quadratic form of $\theta I - B$ is

$$(\theta + a)|u|^2 - 2\operatorname{Re}(\overline{u} c^* v) + v^* D_\theta v.$$

Positivity forces $D_\theta \succeq 0$. Varying a component of v in $\ker D_\theta$ forces $c \perp \ker D_\theta$, equivalent in finite dimension to the range condition. Completing the square on that range leaves $(\theta + a - c^* D_\theta^\dagger c)|u|^2$. This proves necessity and sufficiency. The nonsingular case is the ordinary Schur complement. □

Example 5.4 (An exact local buffer decision)**.** Take a single active pole $s = -2$, parent $a = 1$, and two equally probable child masses $\nu_1 = 0$, $\nu_2 = 4$. The parent mass is two and its Schur sum is one, so its insertion cost is zero, including the zero-root endpoint. The first child already has zero cost at $K = 1$; the second has the exact threshold $K = 2$. Adding any uncoupled positive pole changes full eigenvalue maxima, but none of these insertion-cost conclusions. This is why (5.1) and (5.2) are different certificates.

6 Gap-Free Sensitivity and Certified Choices

The quotient and its residues are exact objects. Directly differentiating root-residue expressions such as $q(\lambda)/p'(\lambda)$ is poorly behaved near collisions. At a fixed deleted polynomial one can instead use the physical arrow, whose change has rank at most two.

Theorem 6.1 (Signed insertion-cost enclosure)**.** *Let $S = S^*$ be a fixed finite Hermitian matrix and*

$$B = \begin{pmatrix} -a & c^* \\ c & S \end{pmatrix}, \qquad \widehat{B} = \begin{pmatrix} -\widehat{a} & \widehat{c}^* \\ \widehat{c} & S \end{pmatrix}, \quad t = \widehat{a} - a, \quad d_c = \|\widehat{c} - c\|_2, \quad D = \sqrt{t^2 + 4d_c^2}.$$

Write $\ell(B) = \operatorname{Tr} B_+ - \operatorname{Tr} S_+$. Then, with no gap, simplicity, or sign hypothesis on a,

$$\frac{-t - D}{2} \;\leq\; \ell(\widehat{B}) - \ell(B) \;\leq\; \frac{-t + D}{2}. \tag{6.1}$$

In particular

$$|\ell(\widehat{B}) - \ell(B)| \le \beta(t, d_c) := \frac{|t| + \sqrt{t^2 + 4d_c^2}}{2}. \tag{6.2}$$

For canonical real nonnegative couplings $c_j = \sqrt{\nu_j}$,

$$d_c^2 = \sum_j (\sqrt{\widehat{\nu}_j} - \sqrt{\nu_j})^2 \le \sum_j |\widehat{\nu}_j - \nu_j|.$$

The constants are independent of the number and spacing of poles.

Proof. For Hermitian X, the variational formula is

$$\operatorname{Tr} X_+ = \max_{0 \preceq P \preceq I} \operatorname{Tr}(PX).$$

For Hermitian Δ it implies

$$-\operatorname{Tr} \Delta_- \le \operatorname{Tr}(X + \Delta)_+ - \operatorname{Tr} X_+ \le \operatorname{Tr} \Delta_+.$$

The matrix $\Delta = \widehat{B} - B$ vanishes outside the span of the first coordinate and $(0, \widehat{c} - c)$. Its possibly nonzero eigenvalues are $(-t \pm D)/2$, one nonnegative and one nonpositive. Insert these in the variational bounds; $\operatorname{Tr} S_+$ cancels exactly. The square-root inequality follows by assuming one mass is larger and expanding the square, or from $|\sqrt{u} - \sqrt{v}| \le \sqrt{|u - v|}$. The zero-difference case follows by continuity and has both bounds zero. □

Scope and meaning of the improvement. The theorem controls the source's actual insertion-cost change, not merely a norm of a formal moment vector. A trace-norm bound would give D in both directions; (6.1) retains the direction of a penalty increase and, when $t = 0$, gives d_c rather than $2d_c$. The main gain is a certified decision margin unaffected by multiple or near-colliding roots. The theorem does not certify residues inferred from inaccurate polynomial coefficients, and it assumes the same physical S at both endpoints. If S changes, the cancellation no longer holds. For aligned equal-size blocks a conservative added error is $2\left\|\widehat{S} - S\right\|_1$, by bounding both positive-part traces, which need not be dimension independent.

Corollary 6.2 (Robust local color selection)**.** *Suppose for each supported child i its exact insertion cost satisfies*

$$\ell_i \in [\widehat{\ell}_i - \beta_i, \widehat{\ell}_i + \beta_i],$$

where arithmetic error in $\widehat{\ell}_i$ is included in β_i. Choose the least index minimizing $U_i = \widehat{\ell}_i + \beta_i$. If all $\beta_i \le \beta$, then

$$\ell_{i_*} \le \min_{i : t_i > 0} \ell_i + 2\beta. \tag{6.3}$$

If a parent has a certified lower cost bound $\ell \ge L_0$ and a certified lower rooted-mass bound $m \ge M_0$, then

$$U_{i_*} \le L_0 + CM_0 \tag{6.4}$$

certifies that child's inequality $\ell_{i_*} \le \ell + Cm$.

Proof. For an exact minimizing child j, $\ell_{i_*} \le U_{i_*} \le U_j \le \ell_j + 2\beta_j$. The last assertion follows from $U_{i_*} \le L_0 + CM_0 \le \ell + Cm$. It uses lower, not upper, parent bounds. □

Failure of (6.4) means "not certified at this precision," not failure of the mathematical pinning inequality. For Borel finite input of fixed size, exact finite minima with least-index tie-breaking and finite arithmetic enclosures are Borel. This elementary observation supplies neither a height-independent selection nor bounds uniform over unbounded slabs.

Proposition 6.3 (Gap-free rooted-mass perturbation)**.** *Let $B, \widehat{B}$ be same-size Hermitian matrices with $\|B\|, \left\|\widehat{B}\right\| \le R$ and $\left\|B - \widehat{B}\right\| \le \delta$, using the same unit root e. For every $\eta > 0$,*

$$|e^* B_+ e - e^* \widehat{B}_+ e| \le \eta + \frac{\delta}{2} + \frac{R\delta}{2\eta}. \tag{6.5}$$

For $R\delta > 0$, choosing $\eta = \sqrt{R\delta/2}$ gives $\delta/2 + \sqrt{2R\delta}$.

Proof. Set $\phi_\eta(x) = (x + \sqrt{x^2 + \eta^2})/2$. Its uniform difference from x_+ is at most $\eta/2$. If $X = (B^2 + \eta^2 I)^{1/2}$ and $Y = (\widehat{B}^2 + \eta^2 I)^{1/2}$, their difference solves

$$X(X - Y) + (X - Y)Y = B^2 - \widehat{B}^2.$$

Since $X, Y \succeq \eta I$, the integral solution of this Sylvester equation gives $\|X - Y\| \le \left\|B^2 - \widehat{B}^2\right\| / (2\eta) \le R\delta/\eta$. Apply the triangle inequality to the two smoothing errors and the difference of ϕ_η values. Minimizing the last two η-dependent terms gives the stated choice. If $R\delta = 0$ the exact difference is zero. □

We do not assert a universal operator-Lipschitz inequality $\left\|B_+ - \widehat{B}_+\right\| \le \left\|B - \widehat{B}\right\|$; the scalar Lipschitz property alone does not justify it.

7 Sharper Deterministic Descent

The source's descent lemma uses a radius-k ball to recover degree-k diagonal moments of a deterministic paved matrix. Closed walks give a smaller sufficient radius. This section concerns only that deterministic stage; it does not change the mixed expected-polynomial locality or the sparse pinning radius in Section 2.

Lemma 7.1 (Half-radius diagonal locality, sharp in general)**.** *Let T be a bounded matrix on the vertices of a bounded-degree graph, with off-diagonal entries supported on its edges. Let v be a vertex, $k \ge 2$, $h = \lfloor k/2 \rfloor$, and T_v its compression to the radius-h ball at v. Then, for every polynomial p of degree at most k,*

$$(p(T))_{vv} = (p(T_v))_{vv}. \tag{7.1}$$

In general the radius cannot be reduced to $h - 1$ while retaining all moments through degree k.

Proof. The vv entry of T^j is a sum over closed walks of length j starting at v; there are only finitely many relevant terms because the graph has bounded degree. If such a walk reaches a point of graph distance d from v, it uses at least d edges to reach it and at least d edges to return, so $2d \le j$. Diagonal steps do not invalidate this inequality. Therefore every contributing walk with $j \le k$ stays in the radius-h ball, proving equality term by term, also for complex Hermitian edge weights. For sharpness use the unit-weight path on vertices $0, \dots, h$, root 0. In degree $2h \le k$, exactly one additional closed walk reaches h and returns. Thus deleting h changes the diagonal moment by one. Rescaling the path adjacency to a contraction preserves a nonzero difference. □

Use the measured setup in Section 2, including the bounded-degree graph, nonsingular relation and cocycle edge bound $J \geq 1$. Let T_{pav} be a deterministic Hermitian contraction after completing a partial base coloring by assigning its unassigned set $\mathcal{U}$ one distinct extra color. Fix $b, \sigma > 0$ and the source buffer $K \geq 1$. On surviving lifted vertices in $[0, L]$, set

$$A_L = d^{1/2}(T_{\mathrm{col}}^L - KbI)d^{1/2}, \qquad m_y = ((A_L)_+)_{yy}, \qquad \Psi = \frac{1}{L}\int m_y \, d\widetilde{\mu}(y),$$

with $m_y = 0$ on deleted vertices. These are now ordinary diagonal spectral costs of a deterministic operator. Let N_h bound the number of vertices in a graph ball of radius h.

Theorem 7.2 (Half-radius and weighted descent)**.** *Suppose p is a real polynomial of degree at most $k \geq 2$ such that*

$$\sup_{[-1,1]} |p(x) - (x - (Kb + \sigma))_+^2| \leq \beta.$$

Put $h = \lfloor k/2 \rfloor$ and assume $L \geq \max\{1, 4h \log J\}$. Then

$$\int_V ((T_{\mathrm{pav}} - (Kb + \sigma)I)_+^2)_{vv} \, d\mu(v) \leq 2\beta + \mu(\mathcal{U}) + \frac{2J^h N_h}{\sigma}\Psi. \tag{7.2}$$

More precisely, let $I = [h \log J, L - h \log J]$, and for surviving lifted y define the finite nonnegative transport weight

$$C_h(y) = \sum_{\substack{u=(v,s),\ v \notin \mathcal{U},\ s \in I \\ \mathrm{dist}(u,y) \leq h}} \frac{d_u}{d_y}, \tag{7.3}$$

where distances are in the original enclosing lifted graph and u, y lie in the same lifted orbit. The last term of (7.2) can be replaced by

$$\frac{1}{\sigma|I|}\int m_y C_h(y) \, d\widetilde{\mu}(y). \tag{7.4}$$

In particular, $C_h(y) \leq J^h N_h$ and $|I| \geq L/2$ recover (7.2). More generally, for $L \geq 1$ and merely $L > 2h \log J$, the weighted estimate still holds and the unweighted last term is

$$\frac{J^h N_h L}{\sigma(L - 2h \log J)}\Psi. \tag{7.5}$$

Proof. Write $g(x) = (x - (Kb + \sigma))_+^2$. Since T_{pav} is a contraction and the threshold is positive, $0 \leq g(T_{\mathrm{pav}}) \leq I$. For each assigned base root v, compress to the assigned vertices in its radius-h ball, giving T_v. The extra color makes every edge between assigned and unassigned vertices zero. Lemma 7.1 therefore gives $(p(T_{\mathrm{pav}}))_{vv} = (p(T_v))_{vv}$. If $\lambda_{\max}(T_v) \leq Kb + \sigma$, the two approximation errors show $(g(T_{\mathrm{pav}}))_{vv} \leq 2\beta$. Let $\mathcal{F}$ be the set of assigned roots where this eigenvalue inequality fails. Bounded finite-ball matrix evaluation makes $\mathcal{F}$ measurable, and

$$\int_V (g(T_{\mathrm{pav}}))_{vv} \, d\mu(v) \leq 2\beta + \mu(\mathcal{U}) + \mu(\mathcal{F}). \tag{7.6}$$

For $u = (v, s)$ with $s \in I$, the radius-h ball lies inside the slab: every edge changes height by at most $\log J$. If Q_u projects onto its assigned lift, $A_L \leq (A_L)_+$ gives

$$T_v - KbI \preceq d_{\mathrm{ball}}^{-1/2} Q_u (A_L)_+ Q_u d_{\mathrm{ball}}^{-1/2}.$$

The positive matrix on the right has trace $\sum_{y\in B_h(u)} m_y/d_y$. When $v \in \mathcal{F}$, its norm and hence its trace exceed σ. Integrate this strict lower bound over $\mathcal{F}\times I$ against $d\mu(v)\,ds$. Replace the set of bad roots by all assigned interior roots on the upper, nonnegative side. The identity $d\mu(v)\,ds = d_u\,d\widetilde{\mu}(u)$ and invariant mass transport yield

$$\sigma|I|\mu(\mathcal{F}) \le \int m_y \sum_{\substack{u \text{ assigned interior}\\ \operatorname{dist}(u,y)\le h}} \frac{d_u}{d_y}\, d\widetilde{\mu}(y).$$

This is the term in (7.4). Each target has at most N_h possible roots, and each ratio is at most J^h. All transports are measurable and integrable on this bounded slab. Since $|I| = L - 2h\log J$, substitution in (7.6) proves (7.5) whenever the interior has positive length, and gives (7.2) under the stronger length assumption. If $J = 1$, $I = [0, L]$ and the same proof applies with no boundary loss. □

Quantitative consequence and boundary. For the same polynomial degree the source factor $J^k N_k$ is replaced by $J^h N_h$, and the required lift margin changes from $k\log J$ to $h\log J$. For degree bound $D \ge 2$, the usual worst-case ball bound is $N_h \le 1 + D\sum_{j=0}^{h-1}(D-1)^j$. On tree-like graphs with $D \ge 3$ this is an exponential reduction of the worst-case ball-volume bound as a function of the approximation degree, not just an adjustment of a universal prefactor. It allows a larger admissible surviving-cost tolerance Ψ at this descent step. It does not change r, the buffer K, the sparse-update recurrence, or the global ε^{-6} color count. The source's degree-k mixed moment locality is left unchanged: a quotient of expected determinants is not the diagonal of the original randomly paved matrix resolvent.

8 Noisy Moments Without False Atomic Recovery

Exact atomic recovery is useful when its algebraic conditions are checked. When moments carry uncertainty, a support-certified polynomial bound is often the honest alternative. Let γ be a probability measure supported on $[-R, R]$, $R > 0$, and suppose

$$|\widehat{\mu}_j - \mu_j| \le e_j, \quad 0 \le j \le n.$$

The support bound is an input; positive Hankel matrices alone do not provide it.

Theorem 8.1 (Explicit one-sided polynomial enclosure)**.** *For $n \ge 1$, set $t = (x+R)/(2R)$ and*

$$P_n(x) = \sum_{j=0}^{n} (2Rj/n - R)_+ \binom{n}{j} t^j (1-t)^{n-j} = \sum_{j=0}^{n} b_j x^j. \tag{8.1}$$

Define $\widehat{M} = \sum_j b_j\widehat{\mu}_j$ and $\eta_m = \sum_j |b_j| e_j$. Then

$$\max\{0, \widehat{M} - \eta_m - R/\sqrt{n}\} \le m \le \widehat{M} + \eta_m. \tag{8.2}$$

This bound requires neither a spectral gap nor an assumed atom count.

Proof. For $J \sim \text{Binomial}(n, t)$, $P_n(x) = \mathbb{E}(2RJ/n - R)_+$. Convexity and $\mathbb{E}(2RJ/n - R) = x$ give $P_n(x) \ge x_+$. The scalar positive-part map is 1-Lipschitz, so

$$P_n(x) - x_+ \le 2R\mathbb{E}|J/n - t| \le 2R\sqrt{t(1-t)/n} \le R/\sqrt{n}.$$

Integrating gives $0 \le L_\mu(P_n) - m \le R/\sqrt{n}$. The moment error is at most η_m by the triangle inequality. Combine this with $m \ge 0$. □

Conversion to monomials may make η_m large through coefficient growth; using a better-conditioned moment bank helps only if its covariance or deterministic error set is transformed too. This is not a claim of numerical stability of high-order monomial recovery. Positivity and interval support can be checked for an exact atomic candidate, or imposed through an independently justified measure model; a finite subset of localizer inequalities is merely necessary unless sufficiency is proved.

The bound is conservative but constructive and local. A faster polynomial approximation can replace (8.1) if its *uniform* error and one-sided ordering are proved. Sampled-grid errors are not uniform certificates. On a fixed slab, source locality says a degree-n rooted moment depends on a finite graph neighborhood. The radius and support bound may deteriorate with the slab, and this theorem supplies no uniform far-deletion estimate.

9 Specific Ram Correspondences and Physical Residuals

The private corpus is used through actual passages, not routing maps. Four correspondences are precise.

Moment coordinates, not a change of problem. TR250's *What a Ram representation means in this report*, equation `eq:moments`, defines a finite measurement map $m = Ec$ and requires a specified inverse on the compatible range. Its `thm:transfer` transports operators and the physical metric together. For the present arrow choose its exact cyclic subspace $X = \operatorname{ran} J$ from Theorem 4.1; it is finite and invariant, and the moments are finite. There are no missing boundary traces. The matrix $H = J^*J$ is the physical metric, not an optional preconditioner.

If $u = Jc$ and an invertible moment bank gives $z = Ec$, put $R_E = E^{-1}$ and

$$C_E = ECE^{-1}, \qquad H_E = R_E^* H R_E. \tag{9.1}$$

Then $H_E C_E = C_E^* H_E$ and $\|JR_E z\|^2 = z^* H_E z$. These identities follow by multiplication and prove exact norm and adjoint transfer. The root vector also changes to Ee_0; retaining e_0 after an arbitrary coordinate change changes the rooted measure. Rectangular redundant coordinates require a declared compatible range and left inverse, not the displayed square inverse.

Hankel positivity, localizers, and an exact annihilator. Volume 21, chapter 02, `finite.tex`, supplies the distinction between feasibility, determinacy, identification, and stability. Here $H_n \succeq 0$ is the Gram matrix of $1, x, \dots, x^n$ in $L^2(\gamma)$; a localizer for $(x+R)(R-x)$ tests a support constraint. Theorem 4.1 uses its `v21c02:zero-energy` and `v21c02:atomic-certificate` mechanism with the actual annihilator p. A small eigenvalue is not an exact annihilator. An incomplete moment bank cannot identify the canceled characteristic factor d at all.

The physical metric detects a missing subspace. Volume 21, `ch01d-physical-metrics.tex`, proves the residual identity for injective synthesis. For a general finite subspace $J : \mathbb{C}^k \to \mathbb{C}^N$, let $H = J^*J$, $P = JH^{-1}J^*$, and $C = H^{-1}J^*BJ$. Then, for every $\lambda \in \mathbb{C}$,

$$\|(B-\lambda)Jc\|^2 = \|(C-\lambda)c\|_H^2 + \|(I-P)BJc\|^2 . \tag{9.2}$$

Indeed the two summands are the orthogonal projections of the residual onto $\operatorname{ran} J$ and its complement. In the exact cyclic construction the second term is zero because $BJ = JC$. If a smaller or numerically estimated subspace is used, it must be measured or bounded; a zero finite residual is not a full spectral certificate. Similarly, $aH \preceq \widehat{H} \preceq bH$, $0 < a \le b$, yields $\sqrt{a}\,\|c\|_H \le \|c\|_{\widehat{H}} \le \sqrt{b}\,\|c\|_H$. Entrywise closeness without a relative lower metric margin is insufficient.

Schur elimination with a residual budget. Core report 5's `prop:schurresidual` requires the full interior residual and its transport to the retained row. Applied to $A_z = zI - B$, $z \in \mathbb{C}^+$, it gives the following particularly simple certificate.

Proposition 9.1 (Inexact rooted resolvent certificate)**.** *Let* $\operatorname{Im} z = \eta > 0$*,* $D = zI - S$*, and let* $(\widehat{u}, \widehat{v})$ *be an approximate solution of* $(zI - B)(u, v) = (1, 0)$*. Define*

$$r_0 = 1 - (z + a)\widehat{u} + c^*\widehat{v}, \qquad r_I = c\widehat{u} - D\widehat{v}.$$

Then

$$1 - h(z)\widehat{u} = r_0 + c^* D^{-1} r_I, \tag{9.3}$$

$$|\widehat{u} - G(z)| \le \frac{|r_0|}{\eta} + \frac{\|c\| \, \|r_I\|}{\eta^2}. \tag{9.4}$$

An alternative full residual bound is $|\widehat{u} - G(z)| \le \|(r_0, r_I)\| / \eta$*.*

Proof. From the second residual equation, $\widehat{v} = D^{-1}(c\widehat{u} - r_I)$. Insert this into the first to obtain (9.3). For Hermitian S, $\|D^{-1}\| \le 1/\eta$; Proposition 3.2 gives $|G(z)| \le 1/\eta$. Multiplying (9.3) by G proves (9.4). The full Hermitian resolvent bound gives the alternative estimate. □

The same core report's polynomial-filter error statement applies to a Hermitian matrix on a certified spectral interval; Theorem 8.1 supplies such a specific polynomial here. Its Woodbury identity can reuse a factored $zI - B$ after a same-space arrow change, but an invertible middle matrix and a full inverse or residual margin are still required. No implementation or measured speedup is claimed for such reuse. Near the real axis the explicit η^{-1} and η^{-2} factors show why a tiny interface residual alone is not persuasive.

10 Verification, Examples, and Limits

The companion finite verification program uses exact rational/complex-rational arithmetic for identities and bounded deterministic numerical experiments for perturbation diagnostics. Its inputs, execution results, source hashes, read-scope records and rendering checks are separate local audit artifacts. Only executed results are to be counted as computational evidence; the proofs above carry the mathematical claims.

10.1 An expected cost is not an expected trace

Take $W = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, zero penalties and two independent uniform colors. With probability $1/2$ the matrix is W, otherwise zero. Thus

$$f(z) = z^2 - \tfrac{1}{2}, \qquad \mathcal{E}(f) = 1/\sqrt{2}, \qquad \mathbb{E} \operatorname{Tr}(W_{\mathrm{pav}})_+ = \tfrac{1}{2}.$$

The two rooted measures each have positive mass cost $1/(2\sqrt{2})$ and sum correctly to $\mathcal{E}(f)$. This counterexample rules out replacing the polynomial cost in the source recurrence by an expected matrix cost.

10.2 Sensitivity at a zero eigenvalue

Let $a = 1$, $S = (-2)$ and $\nu = 2$. The characteristic polynomial is $z(z+3)$ and $\ell = m = 0$. Replacing the residue by $2 + u$, $u > 0$, gives

$$\ell(u) = \frac{\sqrt{9+4u} - 3}{2} > 0, \qquad d_c = \sqrt{2+u} - \sqrt{2}.$$

Theorem 6.1 gives $0 \le \ell(u) \le d_c$ without a gap assumption. The exact zero-cost decision changes at $u = 0$, so interval data spanning both signs cannot certify zero insertion cost. The correct result is an enclosure, not a rounded rank or zero-residue claim.

10.3 Executed checks

The executed exact-arithmetic suite passed 17 check groups using Python 3.10.9, SymPy 1.11.1 and NumPy 1.23.5. It checked six polynomial fixtures of sizes one through four and one through three colors, including Hermitian complex entries, nonuniform probabilities, zero probabilities and an algebraic non-Hermitian complex extension. It checked the differential formula against direct color enumeration, every vertex deletion and every pin average, as well as the square-root determinant representation. The repeated-pole example reproduced $(1, -1, 7, -25, 103)$, its gcd, positive atomic weights, rank-two Hankel matrix, annihilator, physical pencil and resolvent exactly. Six endpoint cases included a singular deleted block and the invisible positive root. Four zero-cost cases, coordinate transfer and the inexact Schur residual identity also passed.

Half-radius moment equality was checked exactly in 36 path/root/degree configurations with degrees two through seven, together with four sharpness fixtures. A separate exact suite checked 88 moment equalities through degrees $k = 2, \dots, 12$ and 11 sharpness fixtures on paths with edge weights $1/3$, hence operator norm at most $2/3$. The change at degree $2h$ upon reducing the radius to $h - 1$ was exactly $(1/3)^{2h}$. These fixtures test the deterministic hypothesis, not a mixed-polynomial radius improvement.

For the atomic example, exact Bernstein integrations at degrees $2, 4, 8, 16$ and deterministic moment errors bounded by 10^{-9} passed their enclosures. At degree 16, the integrated polynomial was exactly $45423251953125/70368744177664$; its noise-adjusted upper bound was approximately 0.64550324, against the exact rooted mass 0.6. The universal lower bound was zero, illustrating that the enclosure need not be sharp. Forty fixed-seed numerical arrow perturbations, with $1, 2, 4, 7, 12$ poles and repeated integer poles allowed, had no violation of the signed cost or smoothed rooted-mass bounds at tolerance 10^{-10}. The smallest signed-cost slack was approximately 0.00460442. Numerical eigensolver tests are diagnostics, not interval proofs. A finite test set does not validate real stability or the measured theorem in arbitrary dimension.

11 Comparison, Unmet Dependencies, and Research

11.1 What has actually been added

Relative to the pinned source, the realized addition is an explicit finite certificate architecture for its actual f/g, with exact treatment of canceled multiplicities and a physical moment pencil. The exact zero-cost threshold can certify a supported color without invoking the very large universal buffer in a qualifying local case. The signed rank-two bound and the parent-lower-bound acceptance rule turn uncertain local computations into explicit admissible or unresolved decisions. The polynomial enclosure provides a fallback that does not presume stable atom recovery. Separately, Theorem 7.2

halves the deterministic moment radius and reduces the slab margin and mass-transport coefficient, while retaining a target-weighted alternative. These are local reformulations and quantitative guarantees, not a new proof of real stability, the MSS bound, or arbitrary-MASA approximation paving.

11.2 Classical tools and private representations

Schur complements, finite spectral measures, cyclic Krylov spaces, Hankel/Vandermonde factorizations, positive-trace variational principles and Bernstein polynomials are classical. The exact coordinate-transfer and metric obligations are developed in the cited private Ram corpus; using them here does not make the underlying results historically new. The source already contains the positive arrow realization and the universal pin inequality. The local contribution is their source-specific certification and sensitivity integration with all normalization and multiplicity obligations retained. No priority claim is made.

The public mixed-determinant and interlacing literature underlies the finite paving method [3, 2]; finite multi-paving can attain quadratic bounds [4]. That fact does not automatically transfer to measurable orbitwise choices. The Popa–Vaes approximation/strong conversion has a normal-expectation hypothesis [5]. The source's expectation-free proof must be judged on its own measured and structural arguments, not inferred from those finite results.

11.3 Advantages and costs

The certificates tolerate repeated poles and root collisions at the level of insertion costs; they keep physical norm accounts visible; and exact algebra distinguishes invisible spectral factors from observable rooted atoms. They can reject an invalid finite input rather than silently changing its interpretation. Conversely, exact characteristic polynomials can be expensive, gcds and root isolation can grow in coefficient complexity, and noisy moment inversion can be ill-conditioned. The method assumes certified residues or a certified physical matrix, not just plausible decimal roots. Polynomial bounds need independent support information and may be loose. None of these constructions supplies a uniform computational complexity bound for a coloring procedure.

11.4 Unmet global obligations

A finite record does not establish a single measurable color at all heights, the uniform far-deletion estimate, sparse-update integrability, or existence of a coloring to which deterministic descent applies. It does not prove the diffuse-kernel structural lemma, analytic modular replacement, concentration for all reduced words, or expected type I decomposition. It does not manufacture a conditional expectation on the original arbitrary MASA or replace the three-corner preservation of off-diagonal blocks. Feldman–Moore's twisted nonsingular representation, foundational free-product constructions and expectations, and standard-form polar-decomposition results remain imported structural theorems rather than independently reproved foundations. These are explicit dependencies of the headline theorem. No contradiction in it is asserted merely because these dependencies are not reproved here.

The most concentrated analytic steps are the high-deletion normal-family passage and the nontracial random-free-moment argument. Their hypotheses and proof dependencies were traced, but no formal or exhaustive verification is claimed. In particular, this report gives no explicit uniform modulus for the source's far-deletion error $e(R)$, no effective slab-length bound, and no effective uniform construction runtime. The large factor $e^{C(S+1)}$ and slab-dependent C_P in the sparse recurrence remain, even after the deterministic descent improvement.

11.5 Research directions with concrete targets

First, derive local finite-radius cost enclosures with a quantitative modulus uniform under the high/low slab deletions, tracking the central moment bound and the Poisson smoothing scale. This would be a genuine bridge beyond the present fixed-S theorem. Second, seek a measured simultaneous two-sided pinning rule to avoid squaring the normalizer color count; finite multi-paving alone is not such a rule. Third, analyze same-space residue extraction from interval polynomial coefficients with multiplicity changes explicitly allowed, rather than assuming a differentiable root labeling. Fourth, compare stable orthogonal-moment banks with monomial banks while measuring the complete physical error budget and arithmetic cost. Any speed or exponent claim should await those proofs and measurements.

A Source Read Scope and Provenance

The source root, relative to `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`, is

> `_sources/openai-math/preprints/Approximation-Paving-over-Arbitrary-Maximal-Abelian-Subalgebras-September-25-2026/build/source/`

All eleven files were read: `main.tex`; `sections/introduction.tex`; `sections/preliminaries.tex`; `sections/polynomials.tex`; `sections/slabs.tex`; `sections/pinning.tex`; `sections/expected.tex`; `sections/expected-atomless.tex`; `sections/general.tex`; `figures/maharam.tex`; and `sources/references.tex`. This includes all proof subsections, the included atomless-kernel argument, the diagram's interpretation, and references. No separate mathematical appendix is present in these eleven files. The figure depicts base fibers, not relation orbits. Full SHA-256 records of these files, the catalogue row, the contribution policy and the actual private files used are retained locally.

The inherited source's bibliography spells the mixed-determinant authors in a different order. The primary public version lists Jonathan Leake and Mohan Ravichandran, which is the order used below. Public checks were retrieval checks on the stated date, not an assertion that a web search exhausts all current literature.

B Exact Private Corpus Passages

All paths below are local and confidential. The mathematical uses are given in Section 9; none of these passages proves the global paving theorem.

1. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Read *Spaces, operator domains, and Ram coordinates*, including *What a Ram representation means in this report*, `eq:moments`; *Ram moment coordinates that retain the complex*, `thm:transfer`; and *A concrete polynomial Ram construction.* These require compatible finite measurement ranges, transported physical metrics and continuity/domain checks, and distinguish coordinate changes from new physical approximation theorems.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch02/finite.tex`. Read the complete file: *What a Moment Record Specifies*; *The polynomial functional*, labels `v21c02:hankel`, `v21c02:localizer`, `v21c02:zero-energy`; *Finite Atomic Recovery With a Complete Check*, `v21c02:atomic-certificate`; the three-point example; and *Finite Nullspaces and Identified Observables.* Its positivity/annihilator proof is the precise finite-measure correspondence used here.

3. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`. Read the complete file, including *A Nonorthogonal Representation Carries a Metric*, `v21c01:physical-gram-definition` and `v21c01:metric-symmetry`; *The Full Residual Has an Omitted Component*, `v21c01:residual-decomposition` and `v21c01:residual-identity`; *Approximating the Metric*, `v21c01:gram-equivalence`; and *Finite Measurements Do Not Give a Continuum Frame*.

4. `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`. Read *Sampling and discarded subspaces*; *Polynomial evaluation without eigendecomposition*, `eq:filtererror`; *Coarsening, Lifts, and Schur Sources*, including *Elimination is not pooling*, `prop:schurresidual`, `eq:schurresidual`; and *Topology Edits and Checked Low-Rank Reuse*, through *Middle conditioning is necessary but not enough*, including `thm:woodbury`. The inexact residual and declared-space requirements, rather than graph-network nomenclature, are the relevant content.

References

[1] OpenAI, *Approximation Paving over Arbitrary Maximal Abelian Subalgebras*, September 25, 2026, pinned local source identified in Appendix A, catalogue M594, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public manuscript.

[2] A. W. Marcus, D. A. Spielman and N. Srivastava, *Interlacing families II: Mixed characteristic polynomials and the Kadison–Singer problem*, Annals of Mathematics **182** (2015), 327–350, especially Theorems 4.1 and 5.1. https://doi.org/10.4007/annals.2015.182.1.8.

[3] J. Leake and M. Ravichandran, *Mixed determinants and the Kadison–Singer problem*, Mathematische Annalen **377** (2020), 511–541; primary version v2. https://arxiv.org/abs/1609.04195v2.

[4] M. Ravichandran and N. Srivastava, *Asymptotically optimal multi-paving*, International Mathematics Research Notices **2021**, 10908–10940, especially Lemma 24 and Theorem 25 in v2. https://arxiv.org/abs/1706.03737v2.

[5] S. Popa and S. Vaes, *Paving over arbitrary MASAs in von Neumann algebras*, Analysis & PDE **8** (2015), 1001–1023, Definitions 2.2–2.3 and Proposition 2.4. https://msp.org/apde/2015/8-4/apde-v8-n4-p06-s.pdf.

[6] M. Aizenman and S. Warzel, *On the ubiquity of the Cauchy distribution in spectral problems*, Probability Theory and Related Fields **163** (2015), 61–87; Boole identity discussion. https://arxiv.org/abs/1312.7769.

[7] M. Subbarao, confidential Ram Calculus corpus passages, exact local paths and sections in Appendix B. These are local representation and certification sources, not public evidence of a new paving theorem.